Prabhavit Gupta

Machine Learning Engineer: GenAI, RAG, Multi-Agent Orchestration & Azure OpenAI

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Summary

Machine Learning and GenAI Engineer with 3 years of commercial experience designing, building, and deploying enterprise-grade LLM applications, advanced RAG architectures, and multi-agent systems. Hands-on experience coordinating intelligent agents via Model Context Protocol (MCP), LangGraph, LangChain, and Semantic Kernel. Proven track record deploying OpenAI models on Azure, implementing high-throughput microservices in Python with FastAPI, and integrating autonomous agents with external APIs, ERPs, and CRMs. Dedicated to engineering rigor with automated PyTest suites, Docker containerization, and production CI/CD pipelines. Permanent EU work authorization (Permesso di Soggiorno UE a Tempo Indeterminato, Italy; fully eligible for remote roles across Spain and Europe).

Experience

FlowSpace.co • *Lead GenAI & Machine Learning Engineer (Freelance)*
Mar 2024 - Aug 2026 • London, UK (remote)

- Architected and deployed production multi-agent systems using Model Context Protocol (MCP) and LangGraph, orchestrating autonomous task execution across enterprise tools and external APIs.
- Designed high-accuracy enterprise RAG pipelines utilizing Azure OpenAI (GPT-4o), hybrid dense/sparse vector retrieval (pgvector), semantic chunking, and cross-encoder reranking.
- Developed modular microservices in Python using FastAPI, incorporating asynchronous message queues and resilient connection handling for 4,500+ daily active users.
- Integrated automated safety guardrails, token rate-limiting, and evaluation harnesses via LangSmith and PyTest, reducing hallucination rates below 1.5%.
- Packaged microservices into production Docker containers, automating CI/CD deployment pipelines on Azure and AWS cloud environments.
- *Tools: Python 3, Azure OpenAI, MCP, LangGraph, LangChain, RAG, FastAPI, Docker, pgvector, PyTest, Git*

Urlink • *GenAI & NLP Engineer (Freelance)*
Apr 2025 - Jul 2025 • Barcelona, Spain (remote)

- Built intelligent document ingestion and classification pipelines utilizing Azure OpenAI embeddings and PostgreSQL/Supabase storage.
- Implemented semantic parsing and structured schema validation layers in Python, ensuring 99.8% extraction accuracy on unstructured business records.
- Designed RESTful API connectors integrating autonomous agent outputs into enterprise CRM and workflow backends.
- *Tools: Python, Azure OpenAI, NLP, REST APIs, PostgreSQL, Supabase, Git, Linux*

Independent Technical Consultant • *AI Solutions Engineer*
Sep 2023 - Mar 2024 • Venice, Italy

- Developed custom conversational AI agents and RAG search assistants for European business clients using LangChain and OpenAI APIs.
- Integrated external SaaS platforms, databases, and webhook listeners to provide automated reasoning and business logic execution.
- Authored technical architecture runbooks, prompt engineering guidelines, and deployment specifications for client engineering teams.
- *Tools: Python, LangChain, OpenAI API, REST APIs, Linux, Git*

Tractionboard.co • *Low-Code Developer (Internship)*
Aug 2023 - Oct 2023 • Utrecht, Netherlands (remote)

- Built automated webhook consumers and data integration workers in Python for SaaS client onboarding flows.
- *Tools: Python, REST APIs, Sendgrid*

Education

Ca' Foscari University of Venice • *Bachelor's in Philosophy, International Studies & Economics, Minor in Entrepreneurship*
Oct 2020 - Jun 2024 • Venice, Italy

Certifications

Microsoft: Introduction to building AI agents • Microsoft Copilot Studio Agent Academy • Microsoft Power Automate • Tray.AI Build Practitioner (Sep 2024) • Tray.AI Build Foundations

Additional Information

- **Work authorization:** Permesso di Soggiorno UE per Soggiornanti di Lungo Periodo (Italy), unrestricted work authorization across the EU (Contrato Indefinido, CCNL, or Partita IVA). No visa sponsorship required.
- **Languages:** English (fluent / C1 professional), Hindi (native), Italian (working intermediate), Spanish (basic comprehension).
- **Availability:** Available immediately for full-time remote roles in Spain and Europe.

Core Skills

- **GenAI & LLM Frameworks:** OpenAI on Azure, Model Context Protocol (MCP), LangGraph, LangChain, Semantic Kernel, Function Calling, Prompt Engineering.
- **RAG & Vector Search:** Retrieval-Augmented Generation, Semantic Search, Hybrid Search, Embeddings, Vector Databases (pgvector, Chroma, Qdrant).
- **Backend & Microservices:** Python 3, FastAPI, REST APIs, Kafka, Asynchronous Programming, SQL (PostgreSQL), NoSQL (MongoDB).
- **Cloud & DevOps:** Azure, AWS, GCP, Databricks, Docker, CI/CD (GitHub Actions, Jenkins), PyTest, Linux.
- **System Integration:** Autonomous Agents, Tool Use, CRM/ERP Integrations, Webhooks, OAuth2, JWT.